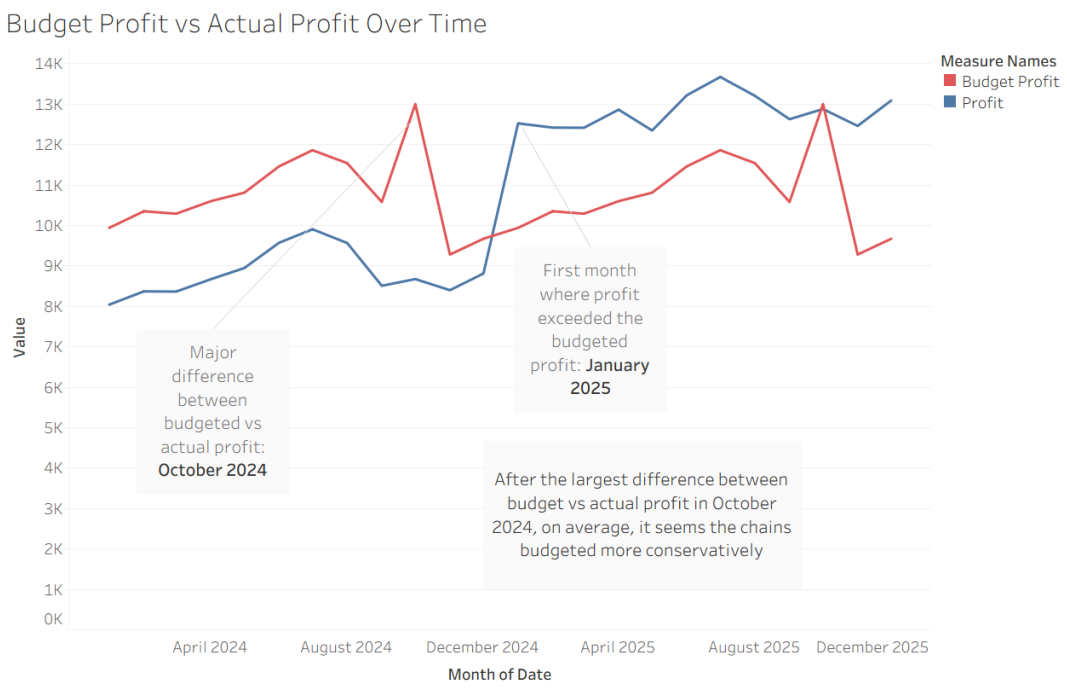
Data Visualization Final Report

Jonathan Magerko

Introduction

This report investigates profitability issues across the coffee chain's products and markets using the dataset provided by IT. The goal is to identify the primary drivers of underperformance and explore factors influencing product profitability.

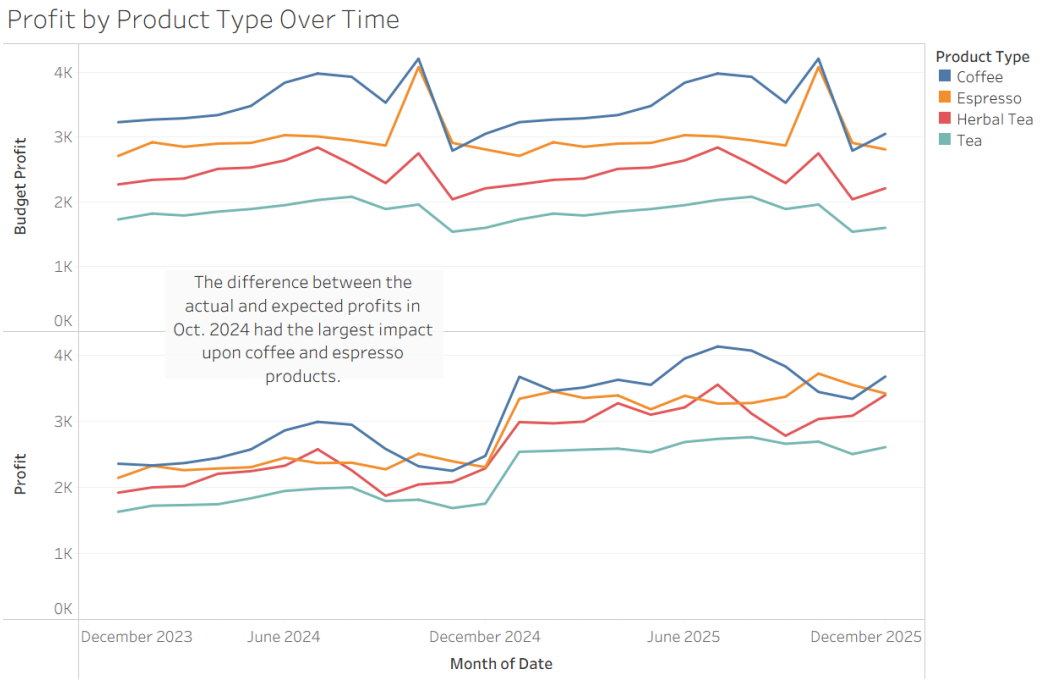
Figure 1: Budget Profit vs. Actual Profit Over Time



I first began by comparing the overall budgeted profit to actual profit over time, as this provides an initial overview of uncertainties in profitability. For all of 2024, budgeted profit exceeded the actual profit. What I believed to be most notable was the drastic difference between

budgeted and actual profit first realized in October 2024. The reason why I believe this was notable is because three months later, in January, actual profit began to exceed the budgeted profit. This may imply that chains began to express more uncertainty and therefore began to expect (or budget for) lower profits. It's important to note however that the spike in budgeted profit occurred the following October, so it's unlikely one event or time period caused profitability issues in general.

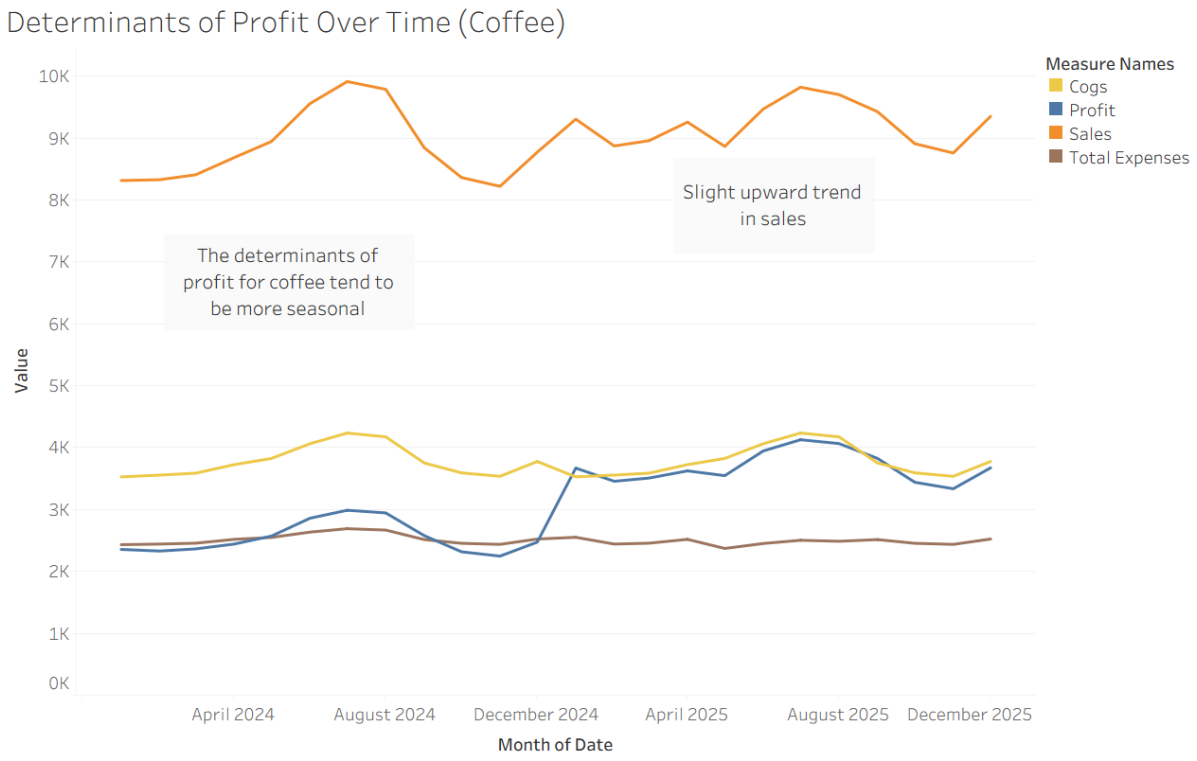
Figure 2: Profit by Product Type Over Time



Even though this October expected profit spike is seasonal, I still wanted to look more in-depth as to how the difference between expected vs. actual profits was distributed amongst the different product types. What I found was that this difference was the greatest amongst coffee

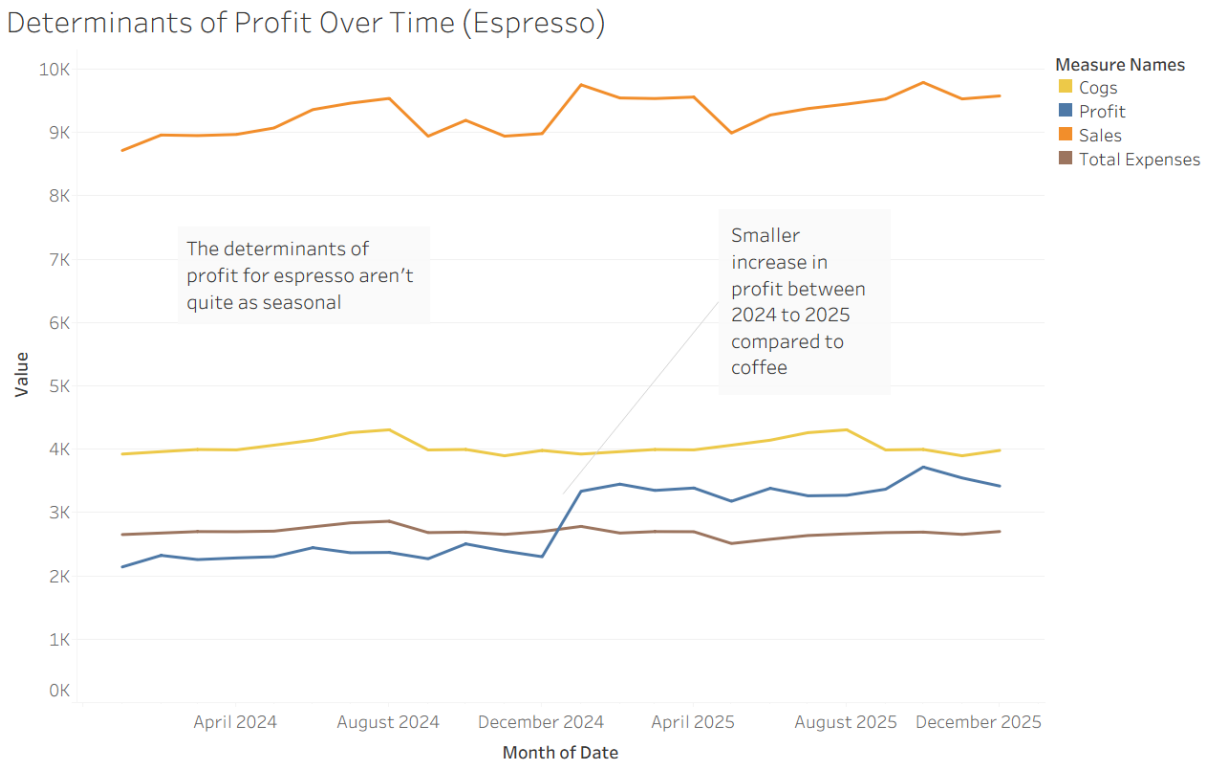
and espresso products, and the effects were consistent when looking at the four different markets individually (Central, East, South, West).

Figure 3: Determinants of Profit Over Time (Coffee)



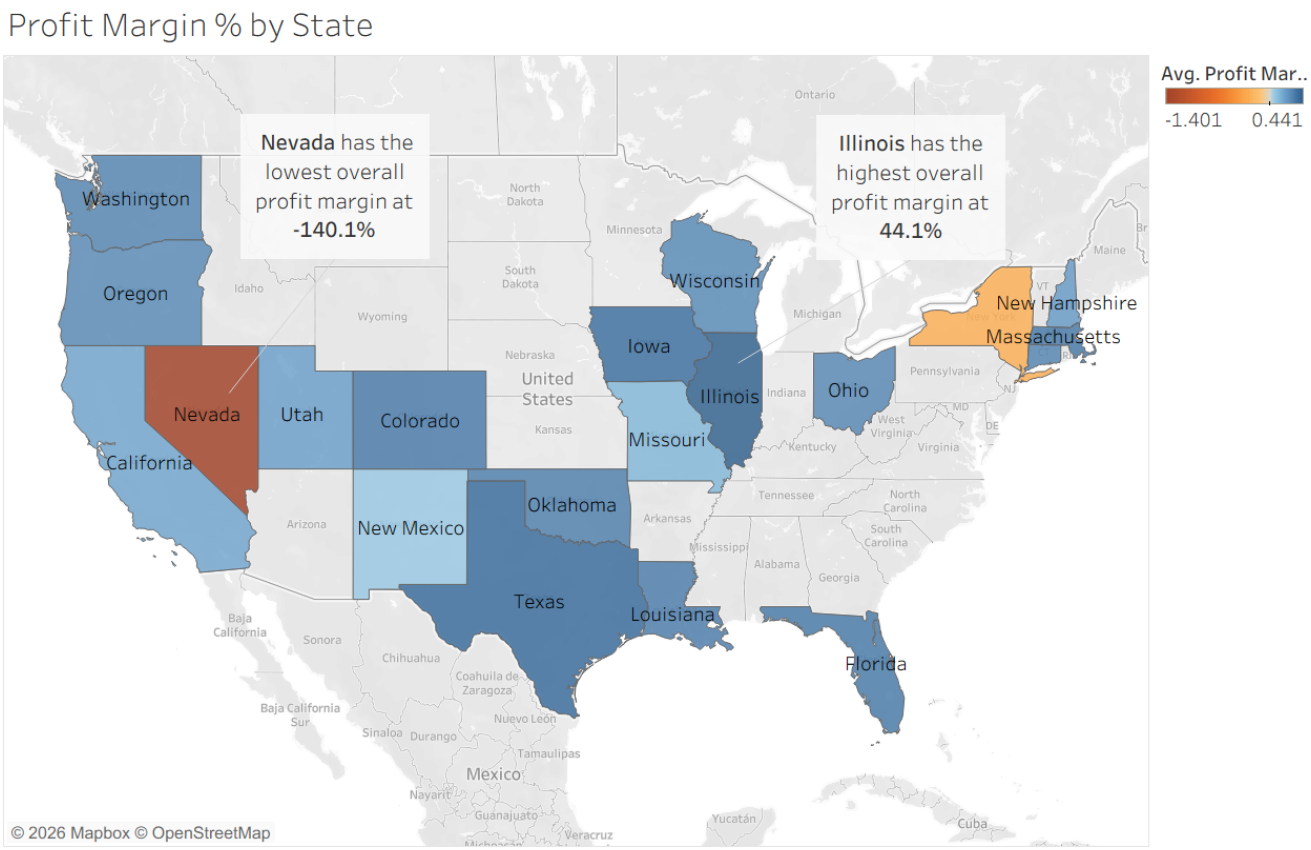
When looking further into how coffee's profit (and its components) varies over time, it was found to be quite seasonal. This seasonality limits the ability to attribute profitability concerns to cost structures, aside from increased uncertainty in forecasting. After accounting for these seasonal fluctuations, a modest upward trend in sales suggests that the increase in profit observed in 2025 may instead be primarily driven by sales growth instead of decreased costs.

Figure 4: Determinants of Profit Over Time (Espresso)



In contrast to coffee, espresso beverages show less seasonality in both profit and its underlying components. While profit in 2025 exceeds that of 2024, the increase is more modest and does not reflect the same magnitude of change observed in coffee. Sales growth appears more steady, and costs remain relatively stable over time. This suggests that espresso contributes to profitability in more of a predictable manner.

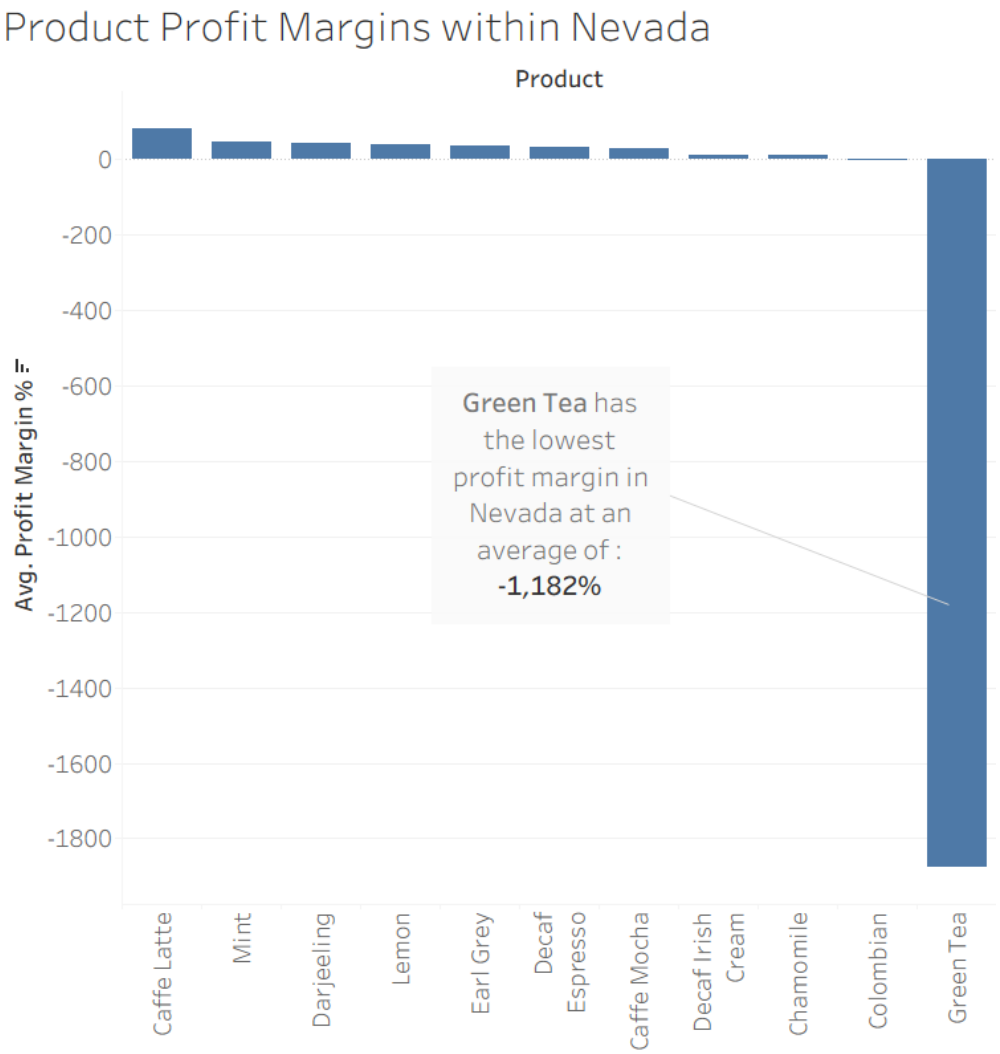
Figure 5: Profit Margin % by State



Since additional product-level detail did not fully explain the observed differences in profitability, I next looked at geographic variation. Rather than comparing total profit across states, I compared profit margins to better assess relative performance. Profit margins allow for a more meaningful comparison by controlling for differences in sales volume across different states.

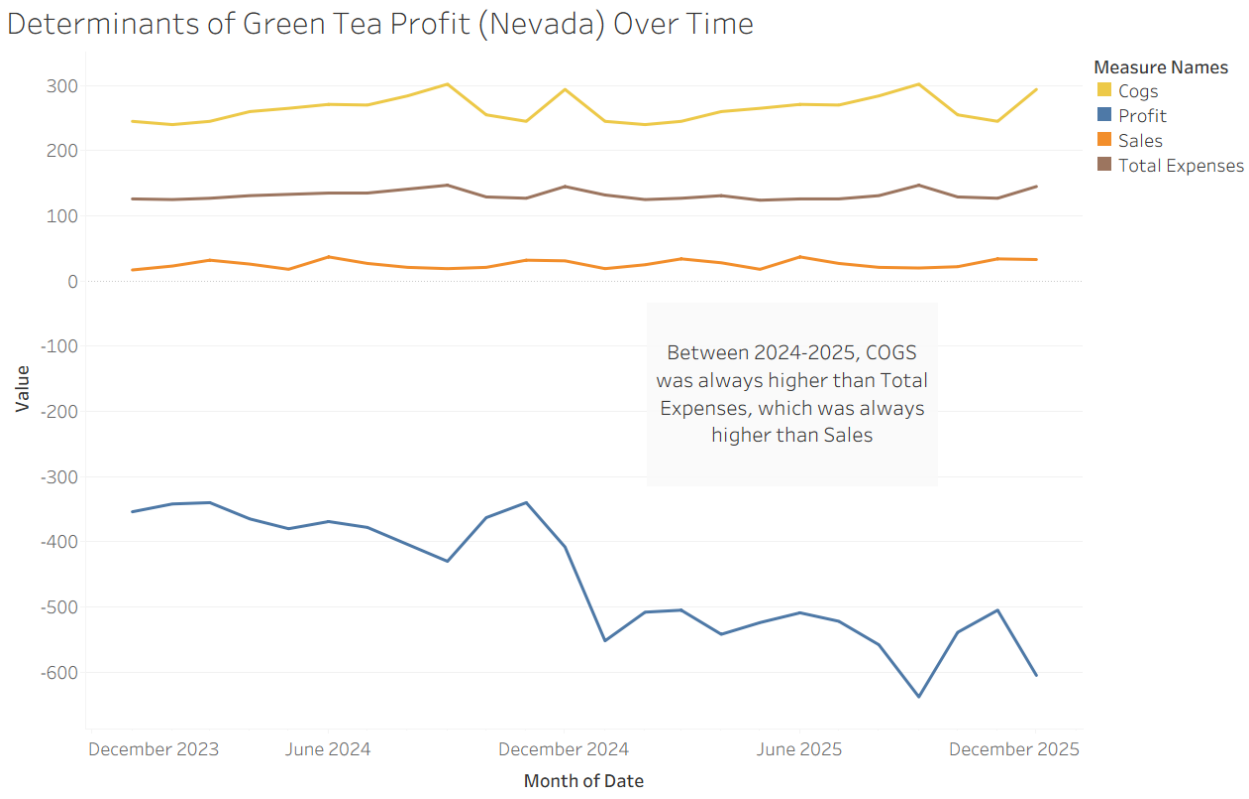
What I found was that most states had relatively consistent profit margins. The median profit margin was 32% with Illinois having the highest at 44.1%. However, Nevada had a considerably different profit margin of -140.1%, suggesting that total expenses in Nevada substantially exceeded sales. This level of underperformance indicates a profitability issue within a state rather than seasonal variation.

Figure 6: Product Profit Margins within Nevada



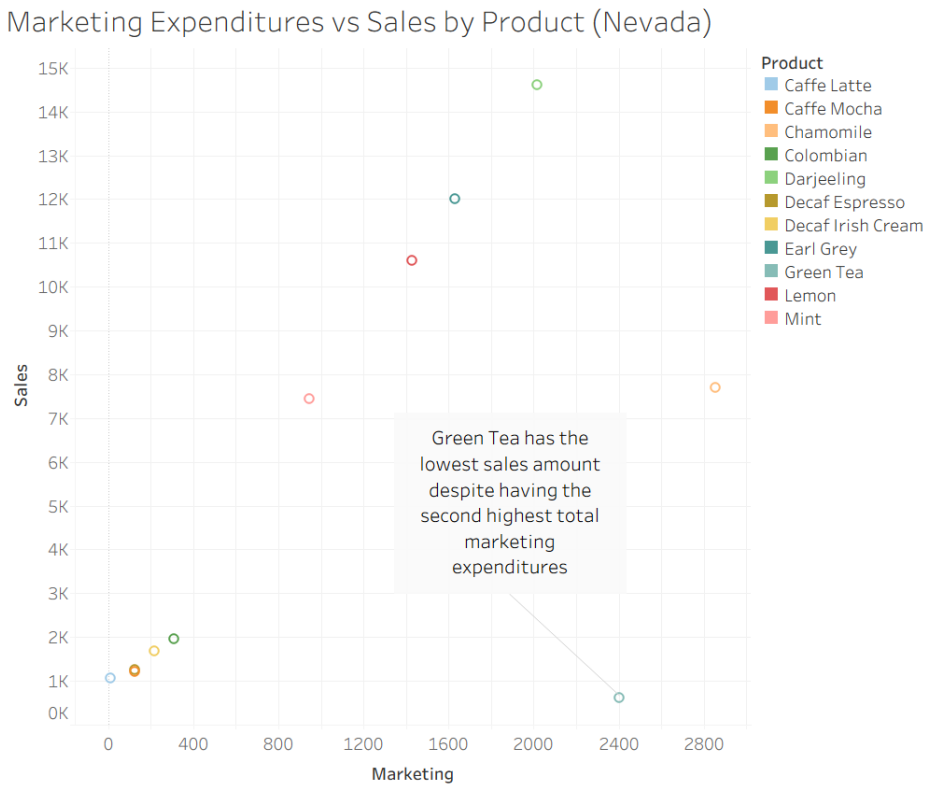
A more thorough analysis of Nevada's profitability revealed that the state's overall negative margin is largely driven by one product category: Green Tea. While most product lines maintain reasonable profit margins within the state, Green Tea has an average profit margin of -1,182%. This suggests that Nevada's underperformance is only concentrated within a certain product line.

Figure 7: Determinants of Green Tea Profit (Nevada) Over Time



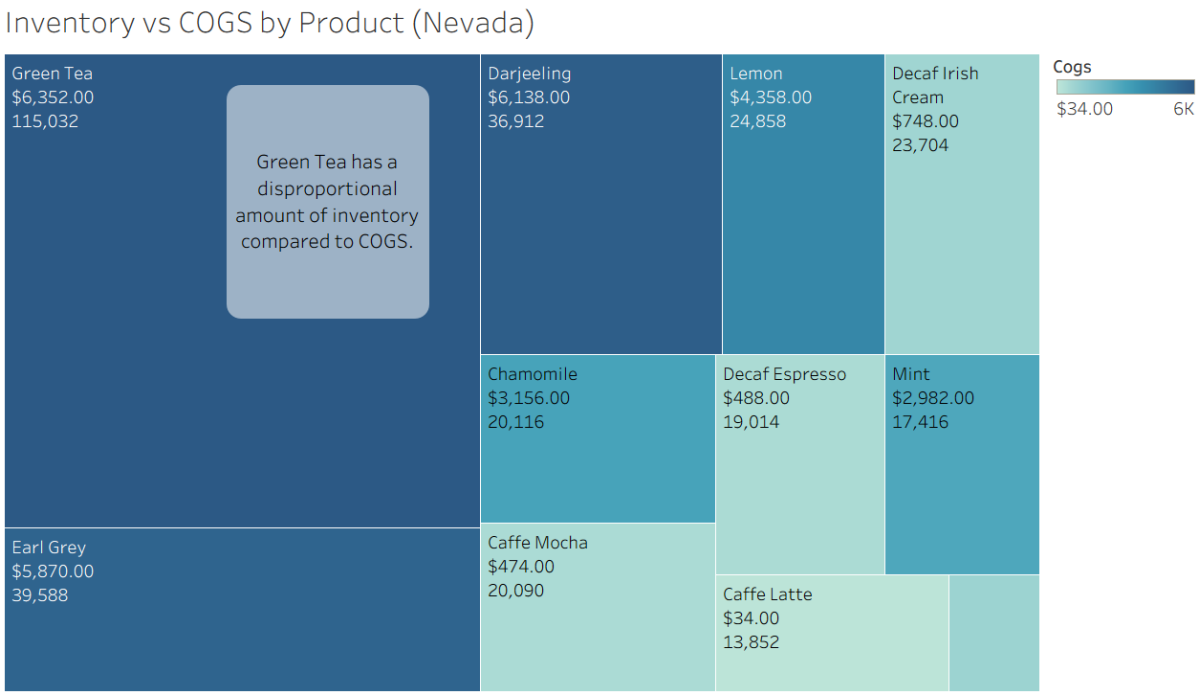
Across both years, costs of goods sold was always greater than the total expenses of the good, and both exceeded the revenue generated from this product. This indicates that the product is unprofitable at the gross level, meaning that either pricing or the costs of production are misaligned with market demand. Since this always has occurred over time, the issue is truly within the Nevada market and is structural.

Figure 8: Marketing Expenditures vs Sales by Product (Nevada)



By comparing marketing expenditures to sales revenue across product types in Nevada, a general pattern was revealed: higher marketing expenditure is correlated with higher sales. (Based on limited number of products) Green Tea however, is a significant outlier, showing the lowest number of sales despite having the second highest marketing expenditure. This reinforces that the product's unprofitability isn't due to insufficient promotion, but rather low demand for the product compared to its high costs.

Figure 9: Inventory vs COGS by Product (Nevada)



This treemap compares inventory levels to cost of goods sold for all products in Nevada. Most products display a relatively similar proportion between inventory and COGS, indicating balanced stocking relative to production costs. Green Tea, however, has a disproportionately high inventory level compared to its COGS. This suggests that Green Tea is overstocked relative to its production costs and sales, essentially tying up limited resources (i.e. space) yet generating insufficient revenue.

Conclusion:

This analysis reveals two primary drivers of profitability across the coffee chain: geographic outliers such as Nevada, and product-level issues such as Green Tea's high costs, low sales, and excessive inventory. Addressing product-level inefficiencies and aligning inventory with sales demand will be important to improve overall profitability.